

lect #4

Prop 2.14 $\exists \mu: \mathcal{P}(\mathbb{R}) \rightarrow [0, +\infty]$ st.

- (i) $\mu(I) = \mathcal{L}(I) \quad \forall I \text{ interval of } \mathbb{R}$
- (ii) $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$ for every disjoint family, $A_n \in \mathcal{P}(\mathbb{R})$
- (iii) $\mu(t+A) = \mu(A) \quad \forall A \subseteq \mathbb{R}, t \in \mathbb{R}$.

Proof (i) - (iii) are all the properties that we used for $(\mathcal{L})^*$ in Prop 2.13, constructing (i) (additivity).

o monotonicity

$$(i) \Rightarrow \mu(\emptyset) = \mathcal{L}(\emptyset) = 0$$

$$A \subseteq B \subseteq \mathbb{R}, \quad B = A \dot{\cup} (B \setminus A) \dot{\cup} \emptyset \dot{\cup} \emptyset \dots$$

$$(ii) \quad \mu(B) = \mu(A) + \underbrace{\mu(B \setminus A)}_{\geq 0} + \underbrace{\mu(\emptyset) + \dots}_{=0}$$

$$\geq \mu(A)$$

o Subadditivity $A_n \subseteq \mathbb{R}$

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A'_n, \quad A'_n = A_n \setminus \left(\bigcup_{j < n} A_j \right)$$

$$\downarrow$$

$$(ii) \Rightarrow \mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \mu\left(\bigcup_{n=1}^{\infty} A'_n\right) \stackrel{\text{pairwise disjoint}}{=} \sum_{n=1}^{\infty} \mu(A'_n)$$

$$\leq \sum_{n=1}^{\infty} \mu(A_n)$$

(monotonicity, just proved)

3. σ -Algebras / Measurable Sets

just saw that for μ^* outer measure e.g., $(\mathcal{L})^*$ it may happen
 $A \cap B = \emptyset \Rightarrow \mu(A \cup B) < \mu(A) + \mu(B)$.

Carathéodory: restrict $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ to a smaller class, which

are the sets $B \subset \mathcal{P}(X)$ st. $\forall A$ s.t. $A \cap B = \emptyset$
 $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$.

(3.1)

Def 3.1 Let $X \neq \emptyset$ $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ outer measure.

A set $A \subseteq X$ is said to be μ^* -measurable if

$$\forall F \subseteq X \quad \mu^*(F) = \mu^*(F \cap A) + \mu^*(F \setminus A). \quad (*)$$

Rmk 3.2 (c) Subadditivity of μ^*

$$\mu^*(F) \leq \mu^*(F \cap A) + \mu^*(F \setminus A) \quad \underline{\text{Always!}}$$

If want to show (*), need to prove

i.e., $\forall F \subseteq X$ $\begin{matrix} \textcircled{\geq} \\ \mu^*(F) \geq \mu^*(F \cap A) + \mu^*(F \setminus A) \end{matrix} \quad (3.2)$

if $\mu^*(F) < +\infty$

(cc) if $\mu^*(E) < +\infty \Rightarrow E$ is μ^* -measurable

let $F \subseteq X$

$$\underbrace{\mu^*(F \cap E)}_{\text{monotonicity} \leq \mu^*(E)} + \underbrace{\mu^*(F \setminus E)}_{\leq \mu^*(F)} \leq \underbrace{\mu^*(E)}_{\leq \mu^*(F)} + \underbrace{\mu^*(F)}_{\leq \mu^*(F)} = \mu^*(F).$$

Corollary 3.3 $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ outer measure

If $B \subseteq X$ is μ^* -measurable. Then $\forall A \subseteq X$, $A \cap B = \emptyset$

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B).$$

Proof B μ^* meas.

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap B) + \mu^*((A \cup B) \setminus B)$$

$$= \mu^*(B) + \mu^*(A) \text{ because } A \cap B = \emptyset. \quad \square$$

Carathéodory Thm (Thm 4.9): the restriction of

$$\mu^*: \mathcal{M}^* \rightarrow [0, +\infty]$$

$$\mathcal{M}^* := \{E \subseteq X, E \text{ } \mu^* \text{-measurable}\} \quad (3.3)$$

then μ^* is countably additive (hence (3.1) holds),

Also, $\mathcal{M}^*$ has important properties, and if $\mathcal{F}$ is what we call a σ -algebra (sigma algebra).

Def 3.4 Let $X \neq \emptyset$, $\mathcal{M} \subseteq \mathcal{P}(X)$ is a σ -algebra if:

$$(i) \quad \emptyset \in \mathcal{M}$$

$$(ii) \quad E \in \mathcal{M} \Rightarrow X \setminus E \in \mathcal{M}$$

$$(iii) \quad E_n \in \mathcal{M}, n \in \mathbb{N}, \text{ then } \bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$$

$(X, \mathcal{M})$ is called a measurable space.

Def 3.5 Let $X \neq \emptyset$, $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ outer measure.

μ^* is said to be regular if:

$$\forall E \subseteq X \quad \exists F \in \mathcal{M}^* \text{ s.t. } E \subseteq F$$

$$\text{and } \mu^*(E) = \mu^*(F).$$

Proposition 3.6 $(X, \mathcal{M})$ measurable space. Then

$$(i) \quad X \in \mathcal{M}$$

$$(ii) \quad E, F \in \mathcal{M} \Rightarrow E \cup F, E \cap F, E \setminus F \in \mathcal{M}$$

$$(iii) \quad E_n \in \mathcal{M}, n \in \mathbb{N} \Rightarrow \bigcap_{n=1}^{\infty} E_n \in \mathcal{M}.$$

Proof (i) $\emptyset \in \mathcal{M}$ (ii) $A \in \mathcal{M} \Rightarrow X = X \setminus \emptyset \in \mathcal{M}$.

$$(ii) \quad E, F \in \mathcal{M} \quad E \cup F = E \cup F \cup \underbrace{\emptyset}_{\in \mathcal{M}} \in \mathcal{M}$$

$$\begin{aligned} E \in \mathcal{M} &\Rightarrow X \setminus E \in \mathcal{M} \\ F \in \mathcal{M} &\Rightarrow X \setminus F \in \mathcal{M} \end{aligned} \quad \Rightarrow (X \setminus E) \cup (X \setminus F) \in \mathcal{M}$$

$$\text{De Morgan's law} \Rightarrow X \setminus (E \cap F) = \underbrace{(X \setminus E)}_{\in \mathcal{M}} \cup \underbrace{(X \setminus F)}_{\in \mathcal{M}} \in \mathcal{M}$$

$$E \cap F = (X \setminus (E \cap F))^c \Rightarrow E \cap F \in \mathcal{M}$$

$$E \setminus F = E \cap (X \setminus F) \in \mathcal{M}$$

$$(iii) \quad E_n \in \mathcal{M} \quad X \setminus \left(\bigcap_{n=1}^{\infty} E_n \right) = \bigcup_{n=1}^{\infty} (X \setminus E_n) \in \mathcal{M}$$

$$\begin{aligned} &\text{measureable} \Leftrightarrow \text{its complement is meas.} \Leftrightarrow \bigcap_{n=1}^{\infty} E_n \in \mathcal{M} \end{aligned}$$

Example 3.7 $X \neq \emptyset$

$$\mathcal{M}_1 := \{\emptyset, X\}, \quad \mathcal{M}_2 := \mathcal{P}(X)$$

$$\mathcal{M} := \{E \subseteq X: E \text{ is countable or } X \setminus E \text{ countable}\}$$

Prop 3.7 $\mathcal{M}_\alpha \subseteq \mathcal{P}(X)$ σ -algebras, $\alpha \in \Lambda$

$$\bigcap_{\alpha} \mathcal{M}_\alpha := \{E \subseteq X: E \in \mathcal{M}_\alpha, \forall \alpha \in \Lambda\}$$

is also a σ -algebra.

Proof Exercises

Def 3.8 let $X \neq \emptyset$, $\mathcal{F} \subseteq \mathcal{P}(X)$.

The σ -algebra generated by $\mathcal{F}$, $\sigma(\mathcal{F})$, is defined as the smallest (wrt to $\subseteq$) σ -algebra that contains $\mathcal{F}$.

$$X = [0,1], \quad \mathcal{F} = \{ [0, 1/2], [3/2, 1] \}$$

Def 3.9 $X = \mathbb{R}^n$ the σ -algebra generated by all open sets of $\mathbb{R}^n$ is called the Borel σ -algebra of $\mathbb{R}^n$.

Denoted $\beta(\mathbb{R}^n)$. Similarly, (X, τ) topological space then $\beta(X) \dots$ σ -algebra generated by all open subsets of X .

Rmk 3.10 F closed $\Leftrightarrow \mathbb{R}^n \setminus F$ open

$$G \in \beta(\mathbb{R}^n) \Rightarrow F = \mathbb{R}^n \setminus G \in \beta(\mathbb{R}^n)$$

$\beta(\mathbb{R}^n) =$ smallest σ -algebra that contains all closed subsets of $\mathbb{R}^n$.

Note that countable sets in $\mathbb{R}^n$ are Borel sets

$$E = \{x_1, x_2, \dots, x_n, \dots\} = \bigcup_{n=1}^{\infty} \{x_n\} \in \beta(\mathbb{R}^n).$$

closed $\in \beta(\mathbb{R}^n)$

Def 3.11 let (X, d) be a metric space. let

$\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ outer measure
 μ^* is said to be a metric outer measure if

$E, F \subseteq X$ if $d(E, F) > 0$ then

$$\mu^*(E \cup F) = \mu^*(E) + \mu^*(F),$$

where $d(E, F) := \inf \{d(x, y) : x \in E, y \in F\}$.

Prop 3.12 If (X, d) is a metric space

$\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ metric outer measure.
Then every Borel set is μ^* -measurable, i.e.,

$$\beta(X) \subseteq \mathcal{M}^*$$
 [see (3.3)].

Prop 3.13 $(\mathbb{R}^n)^*$, H^s_0 , $0 \leq s < +\infty$ [see (2.4)]
(2.2)

are metric outer measures $\Rightarrow$ every Borel set of $\mathbb{R}^n$ is H^s_0 -measurable and $(\mathbb{R}^n)^*$ -measurable.

Proof of Prop 3.12 know [Rmk 3.10]

$$\beta(X) = \sigma(\text{open sets}) = \sigma(\text{closed sets})$$

$$\begin{array}{ccc} \text{So if } \text{closed sets} & \subset & \mathcal{M}^* \\ & \Downarrow & \downarrow \text{r-dgde (stay fixed!)} \\ \beta(X) = \sigma(\text{closed sets}) & \subset & \mathcal{M}^*. \end{array} \quad (4.9)$$

Take $C \subseteq X$ closed ? $C \in \mathcal{M}^*$?

Rmk 3.2: Suffices: $F \subseteq X$, $\mu^*(F) < +\infty$ then (see (3.2))

claim $\cap$. $\mu^*(F \cap C) + \mu^*(F \setminus C) \leq \mu^*(F)$?

For $n \in \mathbb{N}$ define

$$E_n := \{x \in F \setminus C : d(x, C) \geq 1/n\}$$

$$E_n = \{x \in \mathbb{R} \mid C : \frac{1}{n+1} \leq \text{dist}(x, C) < \frac{1}{n}\}$$

E_n are disjoint,

$$\mathbb{R} \setminus C = \bigcup_{n=0}^{\infty} E_n \quad (C \text{ is closed}) \quad (3.4)$$

claim 2 $n \geq 0 \quad \text{dist}(E_{2n}, E_{2n+2}) > 0$

$$x \in E_{2n}, \quad y \in E_{2n+2}$$

$$\frac{1}{2n+1} \leq \text{dist}(x, C) \leq d(x, y) + d(y, C) \leq d(x, y) + \frac{1}{2n+2}$$

$\downarrow x \in E_{2n} \qquad \qquad \qquad \downarrow y \in E_{2n+2}$

$$\Rightarrow d(x, y) \geq \frac{1}{2n+1} - \frac{1}{2n+2} > 0$$

μ^* is a metric outer meas.

$$\mu^*(F) \geq \mu^*\left(\bigcup_{n=0}^k E_{2n}\right) = \sum_{n=0}^k \mu^*(E_{2n}) \quad \forall k$$

$$\Downarrow \sum_{n=0}^k \mu^*(E_{2n}) \leq \mu^*(F)$$

$$\Rightarrow \sum_{n=0}^{\infty} \mu^*(E_{2n}) \leq \mu^*(F)$$

Similarly, $\sum_{n=1}^{\infty} \mu^*(E_{2n-1})$ conv.

$$\Rightarrow \left[\sum_{n=0}^{\infty} \mu^*(E_n) \right] \text{ converges.} \quad (3.5)$$

claim 3

$$\forall k \in \mathbb{N}, \quad \text{dist}\left(F \setminus C, \bigcup_{n=0}^k E_n\right) > 0$$

$$x \in F \setminus C, \quad y \in \bigcup_{n=0}^k E_n$$

$$\frac{1}{k+1} \leq d(y, C) \leq d(x, y)$$

μ^* metric outer measure:

$$\mu^*(F \cap C) + \mu^*(F \cap C^c) = \mu^*(F \cap C) + \mu^*\left(\bigcup_{n=0}^{\infty} E_n\right)$$

$$= \mu^*(F \cap C) + \mu^*\left(\bigcup_{n=0}^k E_n \cup \bigcup_{n=k+1}^{\infty} E_n\right)$$

subd.

$$\leq \mu^*(F \cap C) + \mu^*\left(\bigcup_{n=0}^k E_n\right) + \mu^*\left(\bigcup_{n=k+1}^{\infty} E_n\right)$$

(claim 3)

$$= \mu^*\left(\underbrace{(F \cap C) \cup \left(\bigcup_{n=0}^k E_n\right)}_{\subseteq F}\right) + \sum_{n=k+1}^{\infty} \mu^*(E_n)$$

$$\leq \sum_{n=k+1}^{\infty} \mu^*(E_n)$$

$$\text{mon.} \quad \leq \mu^*(F) + \sum_{n=k+1}^{\infty} \mu^*(E_n) \quad \forall k \geq 0$$

$$\downarrow_0 \text{ as } k \rightarrow \infty$$

let $k \rightarrow \infty$ and get

$$\mu^*(F \cap C) + \mu^*(F \cap C^c) \leq \mu^*(F).$$